Extra Dessert #20: Device and Circuit Symbol Drill

Every schematic symbol you may be asked to identify, with the rule that distinguishes it from its trap-answer cousins.

Supports EC-B and all E6/E7 Bites · Cover the right column and quiz yourself · Self-study reference · Practice exams at eham.net/exams

How To Use This Sheet

This is a recognition drill. Cover the right column with a piece of paper and try to name the device from the symbol description and identifying cue. Uncover, check, repeat. The goal is not memorizing pictures – it is getting the **cue** into your head so you can name any symbol the exam puts in front of you.

The pool's symbol-ID questions are concentrated in three official figures:

- **Figure E6-1** – six FET symbols. Tests **symbol 1 (P-channel JFET)** and **symbol 4 (N-channel dual-gate MOSFET)**. Cues: arrow on gate direction, count of gate lines.
- **Figure E6-2** – six diode symbols. Tests **symbol 6 (Schottky diode)**. Cue: distinctive "S"-hook on the cathode bar.
- **Figure E7-1** – a single-transistor amplifier circuit. Tests R1/R2 = **voltage divider bias**, R3 = **self bias**, and the circuit type = **common emitter**.

Passive Components

Symbol	Name	Distinguishing cue
Zigzag line	Resistor (R)	Zigzag is US style. International style is a plain rectangle. Either way: ohms of constant resistance, no polarity.
Two parallel plates	Capacitor, non-polarized (C)	Two equal-length plates. Blocks DC, passes AC.
Plate and curved plate	Capacitor, polarized (electrolytic)	One straight plate, one curved plate (or "+" mark). Big capacitance values, polarity matters.
Coil or series of arcs	Inductor (L)	Loops or arcs in a line. Passes DC, blocks high-frequency AC.
Coil with parallel lines beside	Inductor with iron/ferrite core	The straight lines next to the coil indicate the core type.
Pair of coils facing each other	Transformer (T)	Two coils sharing a core. Lines between them = laminated iron core; no lines = air core.

Diodes And Special Diodes

Symbol	Name	Distinguishing cue
Triangle pointing into a bar	Standard diode (D)	Triangle = anode, bar = cathode. Current flows in the direction the triangle points.
Diode with "Z" or stepped cathode bar	Zener diode	Cathode bar bent into a "Z" shape. Voltage reference; clamps at a precise reverse voltage.
Diode with "S" or hook cathode bar	**Schottky diode (Fig E6-2 sym 6)**	Cathode bar has an "S"-curve hook. Metal-semiconductor junction; low V drop; fast.
Diode with two-line cathode (capacitor-like)	Varactor diode	Cathode looks like a small capacitor plate.

		Voltage-controlled capacitance.
Diode with arrows pointing in	Photodiode / phototransistor	Inward arrows = light coming in.
Diode with arrows pointing out	LED	Outward arrows = light going out.

Transistors And FETs

Symbol	Name	Distinguishing cue
Circle with B, C, E; arrow on E pointing **out**	NPN bipolar transistor (Q)	Arrow leaves the emitter. “Not Pointing iN.”
Circle with B, C, E; arrow on E pointing **in**	PNP bipolar transistor (Q)	Arrow points into the emitter. “Pointing iN to body.”
G, D, S with arrow on gate pointing **in**	N-channel JFET	Single gate line with arrow into the channel. **N for In.**
G, D, S with arrow on gate pointing **out**	**P-channel JFET (Fig E6-1 sym 1)**	Single gate line, arrow points away from the channel.
G1, G2, D, S; arrow on channel into	**N-channel dual-gate MOSFET (Fig E6-1 sym 4)**	**Two** gate inputs labeled G1 and G2; channel arrow points in.
G1, G2, D, S; arrow on channel out	P-channel dual-gate MOSFET	Two gates; channel arrow points out (P-channel).
G, D, S with solid channel line	Depletion-mode FET	Channel line is **solid**. Conducts at zero gate voltage.
G, D, S with dashed/ broken channel line	Enhancement-mode FET	Channel line is **broken**. No channel until gate is driven.

Common Bias And Configuration Cues (Figure E7-1)

Cue in the schematic	What it means
Two resistors forming a divider from supply to ground, tap goes to base	**Voltage divider bias** (R1 + R2 in Fig E7-1). Sets a stable DC base voltage.
Resistor between emitter and ground (often with a bypass cap across it)	**Self bias / emitter degeneration** (R3 in Fig E7-1). Stabilizes operating point against beta variation.
Capacitor across the emitter resistor	**Emitter bypass** – shorts the emitter resistor at signal frequencies to preserve AC gain.
Input couples to base, output taken from collector, emitter at AC ground	**Common-emitter** amplifier. Voltage and current gain; output phase-inverted from input.
Input couples to base, output taken from emitter	**Common-collector / emitter follower**. Voltage gain ≈ 1, current gain high, output in phase with input.
Input couples to emitter, base grounded for AC	**Common-base / grounded-grid**. Low input impedance.

Grounding And Wire Junctions

Symbol	Name	Distinguishing cue
Triangle/stack pointing down	Ground (GND)	The zero-volt reference. All ground symbols connect together electrically.

Filled black dot at a crossing	Junction	Wires that cross **are** connected.
No dot at a crossing	Crossover	Wires that cross are **not** connected. Beginners miss this constantly – check for the dot.

Speed-recognition discipline: on the exam you have time to read each diagram carefully, but on these symbol questions you should be able to name the device in 5 seconds or less. If a symbol slows you down, that is a “Weak” mark on your D19 score tracker – come back and drill it again.